

INTRODUCTION TO WEB DEVELOPMENT

LEARNING OBJECTIVES

- Define web development
- Identify and learn the foundations of front-end development
- Review the fundamentals of HTML and CSS

Old Website



Welcome to My Homepage!

**Under Construction**



[Home](#)[About Me](#)[My Links](#)[My Photos](#)[Guestbook](#)

Welcome to My Website!

Latest Updates:

- New! Check out my [photo gallery](#)!
- Added some cool animated GIFs!
- Sign my [guestbook](#)!

A Cool Animated GIF:



Visit My Links:

[Visit My Links](#) Page for some great sites!

VS

Modern Website



Creative Digital Agency

We Build Amazing Websites & Apps

**Web Design****SEO Marketing****Mobile Apps**



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We create innovative solutions to help your business grow online. Learn more about our team and our story.

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**15**
Years of Experience

**300+**
Happy Clients

**120+**
Projects Completed

WHAT IS WEB DEVELOPMENT?

Web development refers to building, creating, and maintaining websites. It includes aspects such as web design, web publishing, web programming, and database management. It is the creation of an application that works over the internet, i.e., websites.

WHAT IS WEB DEVELOPMENT?

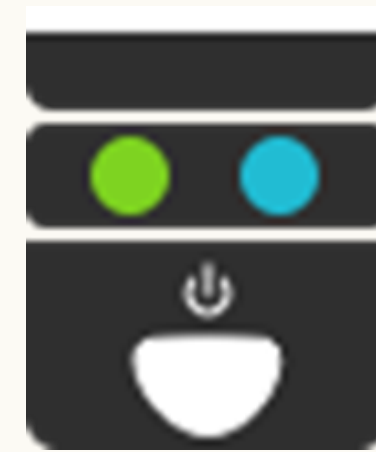
Developers write code that describes...

What is shown
in the browser



“front-end development”
or “client-side code”

What should be sent to
the browser (and more)



“back-end development”
or “server-side code”

WHAT IS WEB DEVELOPMENT?

There are two major areas: **Frontend** and **Backend** which forms the backbone of web development. Each plays a crucial role in creating seamless, functional web experiences.

WHAT IS WEB DEVELOPMENT?

- It involves frontend (UI/UX), backend (server-side logic), and databases.
- Developers use programming languages, frameworks, and tools to build websites.

FRONTEND VS BACKEND VS FULL-STACK

- **Frontend Development:** User interface and experience (HTML, CSS, JavaScript).
- **Backend Development:** Server-side logic, databases, APIs (Node.js, Python, PHP, etc.).
- **Full-Stack Development:** Both frontend and backend expertise.

CAREER PATHS IN WEB DEVELOPMENT

- Frontend Developer
- Backend Developer
- Full-Stack Developer
- UI/UX Designer
- DevOps Engineer
- Web Security Specialist

FRONT-END

The foundations of front-end development are...

HTML



CSS



JS



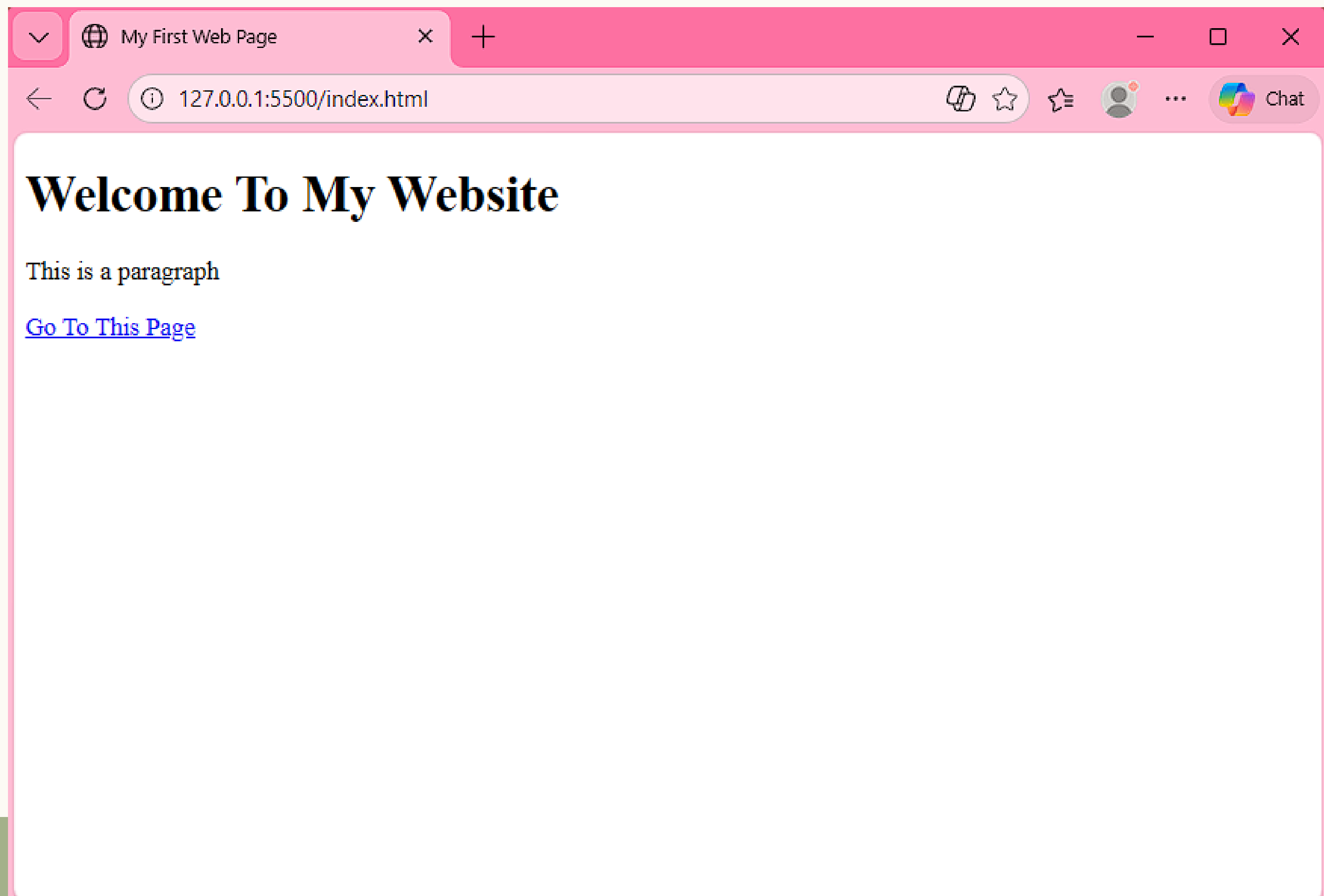
WHAT IS HTML?

HYPertext MARKUP LANGUAGE

HTML is the standard **markup language** used to create the structure of web pages. It consists of elements (or tags) that define different parts of a webpage, such as headings, paragraphs, images, links, and forms.

HYPERTEXT MARKUP LANGUAGE

- **Hypertext** refers to the pieces of text which can link to other documents in the website.
- **Markup Language** specifies the structure and formatting of a document and can also define the relationships among its parts.



HYPER TEXT MARKUP LANGUAGE

- **Tags**
- **Elements**
- **Attributes**

HYPER TEXT MARKUP LANGUAGE

- **Tags** – are special keywords enclosed in angle brackets that act as the fundamental building blocks of a webpage.
- **Elements** – composed of start tag, content, and an end tag. It tell the browser how to display the content
- **Attributes** – special words used within the opening tag of an HTML element to provide additional information or modify its default behavior and appearance.

HYPER TEXT MARKUP LANGUAGE

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title> </title>
```

```
</head>
```

```
<body>
```

```
</body>
```

```
</html>
```

HYPER TEXT MARKUP LANGUAGE

<!DOCTYPE html> - declaration defines that this document is an HTML5 document

<html> - the root element of an HTML page

<head> - contains meta information about the HTML page

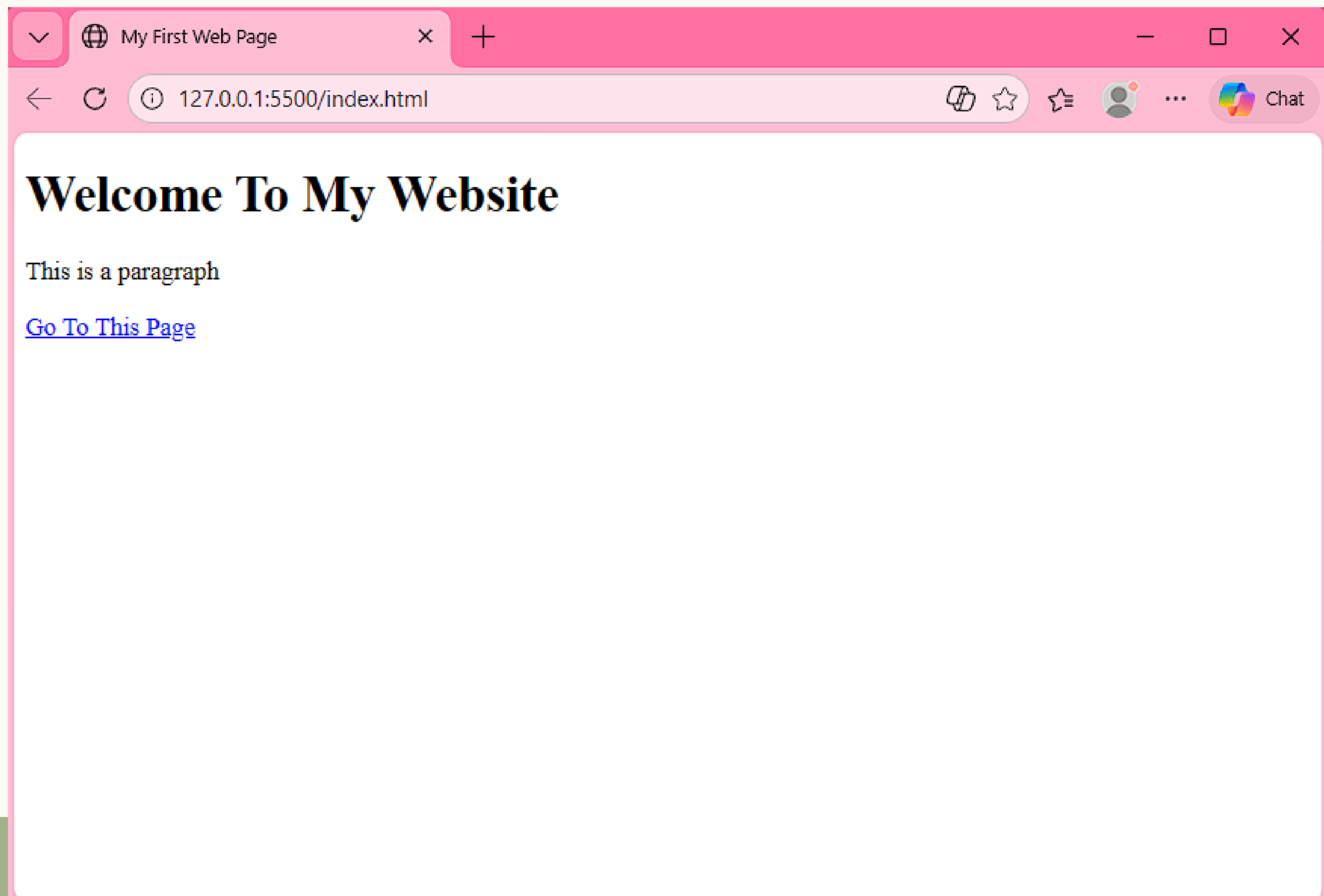
<title> </title> - specifies a title for the HTML page

</head>

<body> - defines the document's body, and is a container for all the visible contents

</body>

</html>



index.html > ...

```
1  <!DOCTYPE html>
2  <html>
3
4  <head>
5  |   <title>My First Web Page</title>
6  </head>
7
8  <body>
9  |   <h1>Welcome To My Website</h1>
10 |   <p>This is a paragraph</p>
11 |   <a href="https://www.w3schools.com/html/default.asp">Go To This Page</a>
12 </body>
13
14 </html>
```

HTML BASIC TAGS

- HTML **headings** are defined with the `<h1>` to `<h6>` tags.
- HTML **paragraphs** are defined with the `<p>` tag.
- HTML **links** are defined with the `<a>` tag.
- HTML **images** are defined with the `<img>` tag.
- The `<br>` tag defines a **line break** and is an empty element without a closing tag.

HTML BASIC TAGS

- The `<hr>` tag defines a **thematic break** in an HTML page.
- The HTML `<pre>` element defines preformatted text.
- `<!-- Write your comments here -->`

HTML BASIC ATTRIBUTES

- href for `<a>`
- src for `<img>`
- width and height for `<img>`
- alt for `<img>`
- style
- lang for `<html>`
- title

HTML TEXT FORMATTING

- `<b>` - Bold text
- `<strong>` - Important text
- `<i>` - Italic text
- `<em>` - Emphasized text
- `<mark>` - Marked text
- `<small>` - Smaller text
- `<del>` - Deleted text
- `<ins>` - Inserted text
- `<sub>` - Subscript text
- `<sup>` - Superscript text

HTML LINKS

The **target** attribute can have one of the following values:

- **_self** – Default. Opens the document in the same window/tab as it was clicked
- **_blank** – Opens the document in a new window or tab
- **_parent** – Opens the document in the parent frame
- **_top** – Opens the document in the full body of the window

HTML LINKS

- To use an image as a link, just put the `<img>` tag inside the `<a>` tag
- Use **mailto:** inside the **href** attribute to create a link that opens the user's email program (to let them send a new email)

HTML LISTS

- ORDERED LIST `<ol>`
- UNORDERED LIST `<ul>`
- DESCRIPTION LIST `<dl>`

ORDERED LIST

- ORDERED LIST – Items are numbered.

`<ol>` - ordered list

`<li>` - list of item

type - an attribute that defines the type of the list item marker.

start - start counting from a specified number.

UNORDERED LIST

- UNORDERED LIST – Rendered as bulleted list.

`<ul>` - unordered list(bulleted)

`<li>` - list of item

CSS list-style-type property - used to define the style of the list item marker.

DESCRIPTION LIST

- DESCRIPTION LIST – used for terms and descriptions

`<dl>` - description list

`<dt>` - description term

`<dd>` - describe the term in a description list

NESTING AND INDENTATION

- Important concepts in HTML that help organize and structure your code, making it more readable and maintainable.
- NESTING – refers to placing HTML elements inside other elements.
- INDENTATION – the practice of adding spaces or tabs at the beginning of all lines of code to visually represent the nesting structure

SEMANTIC ELEMENTS

- The semantic elements are new with HTML5, but they have become very important for Search Engine Optimization (SEO).
- The semantic elements describe the structure of the content for the browser.

header Element

```
<header></header>
```

nav Element

```
<nav></nav>
```

main Element

```
<main></main>
```

footer Element

```
<footer></footer>
```

SEMANTIC ELEMENTS

`<header></header>`

- represents a container for introductory content or a set of navigational links.

header Element

```
<header></header>
```

nav Element

```
<nav></nav>
```

main Element

```
<main></main>
```

footer Element

```
<footer></footer>
```

SEMANTIC ELEMENTS

`<nav></nav>`

- defines a set of navigation links.
- Notice that NOT all links of a document should be inside a `<nav>` element. The `<nav>` element is intended only for major blocks of navigation links.

header Element

```
<header></header>
```

nav Element

```
<nav></nav>
```

main Element

```
<main></main>
```

footer Element

```
<footer></footer>
```

SEMANTIC ELEMENTS

`<main></main>`

- specifies the main content of a document.
- should be unique to the document.
- There must not be more than one `<main>` element in a document.

header Element

```
<header></header>
```

nav Element

```
<nav></nav>
```

main Element

```
<main></main>
```

footer Element

```
<footer></footer>
```


SEMANTIC ELEMENTS

<footer></footer>

defines a footer for a document or section.

A **<footer>** element typically contains:

- authorship information
- copyright information
- contact information
- sitemap
- back to top links
- related documents

header Element

```
<header></header>
```

nav Element

```
<nav></nav>
```

main Element

```
<main></main>
```

footer Element

```
<footer></footer>
```

<header>

<nav>

<main>

<footer>



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HTML/CSS ▾

WEB DESIGN

PHP/MYSQL

BEAVER BUILDER

BLOG

[Home](#) » [Sample Web Page](#)

Sample Web Page

[Edit](#)

This is the text on a sample web page

Other Websites

[WhiteBuffaloWebsites.com](#)

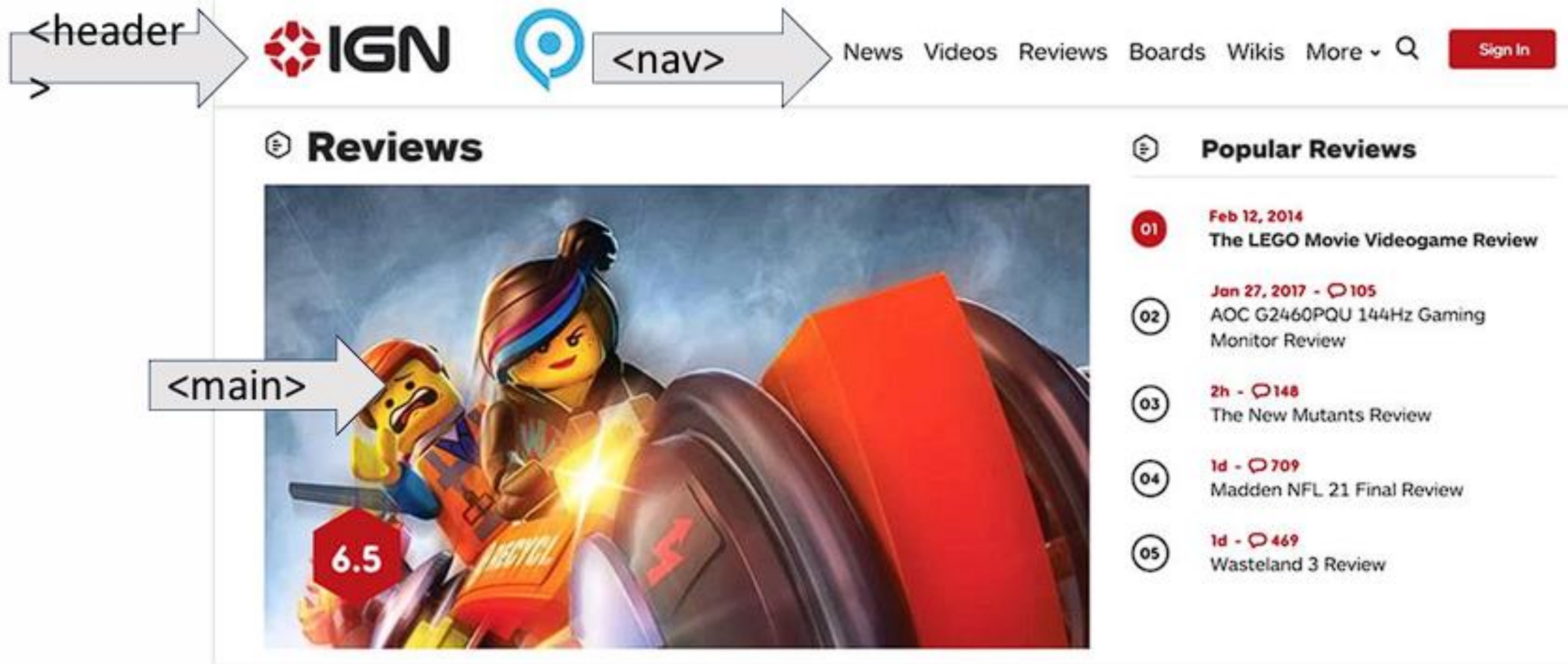
[WhiteBuffaloKids.com](#)

[HatsOffAmerica.com](#)

[SusanMetoxen.com](#)

Note that the header, nav and footer will be the same on every page. The main section will be different on every page.

Web Dev Students is a product of White Buffalo LLC and Susan Metoxen. Susan Metoxen is a part-time computer science instructor at [Saint Paul College](#) and the owner of [White Buffalo Websites](#).
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This is another common layout, with the `<nav>` next to (or inside) or the `<header>`

NON-SEMANTIC ELEMENTS

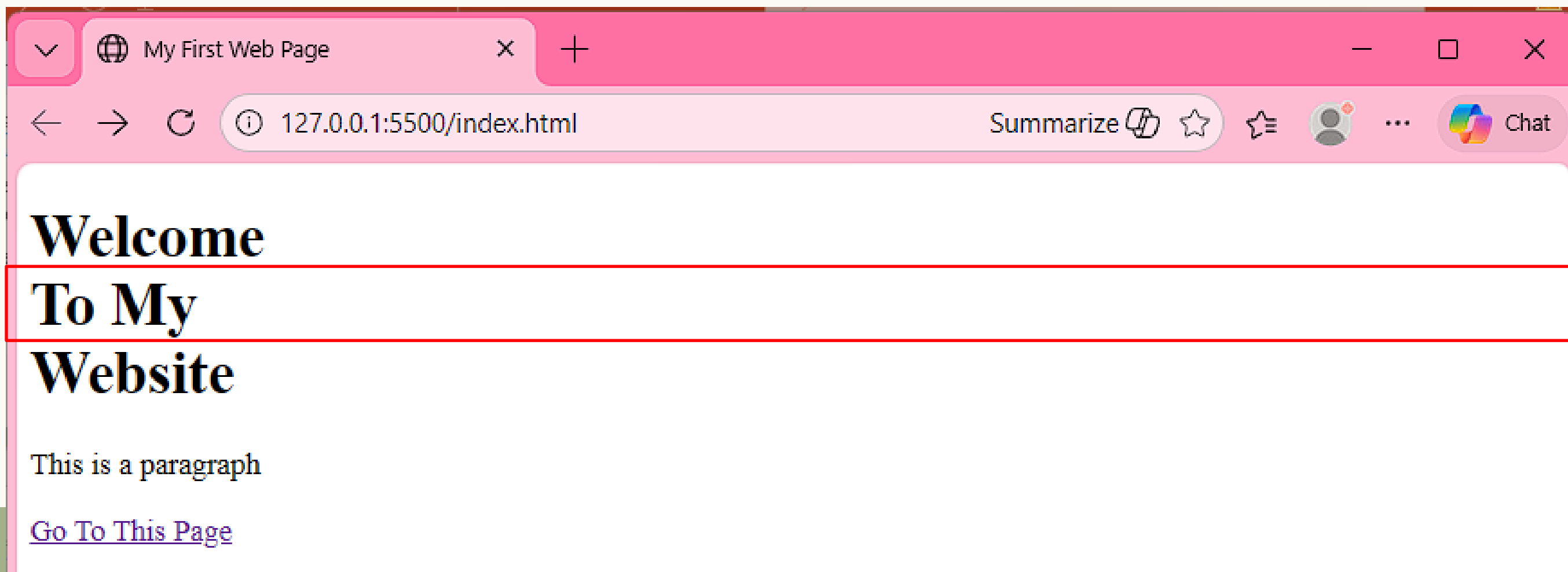
- Tags that do not provide any information about the content they contain to the browser or developer beyond their basic structural type.
- They are primarily used as generic containers for styling and layout purposes.

NON-SEMANTIC ELEMENTS

`<div></div>`

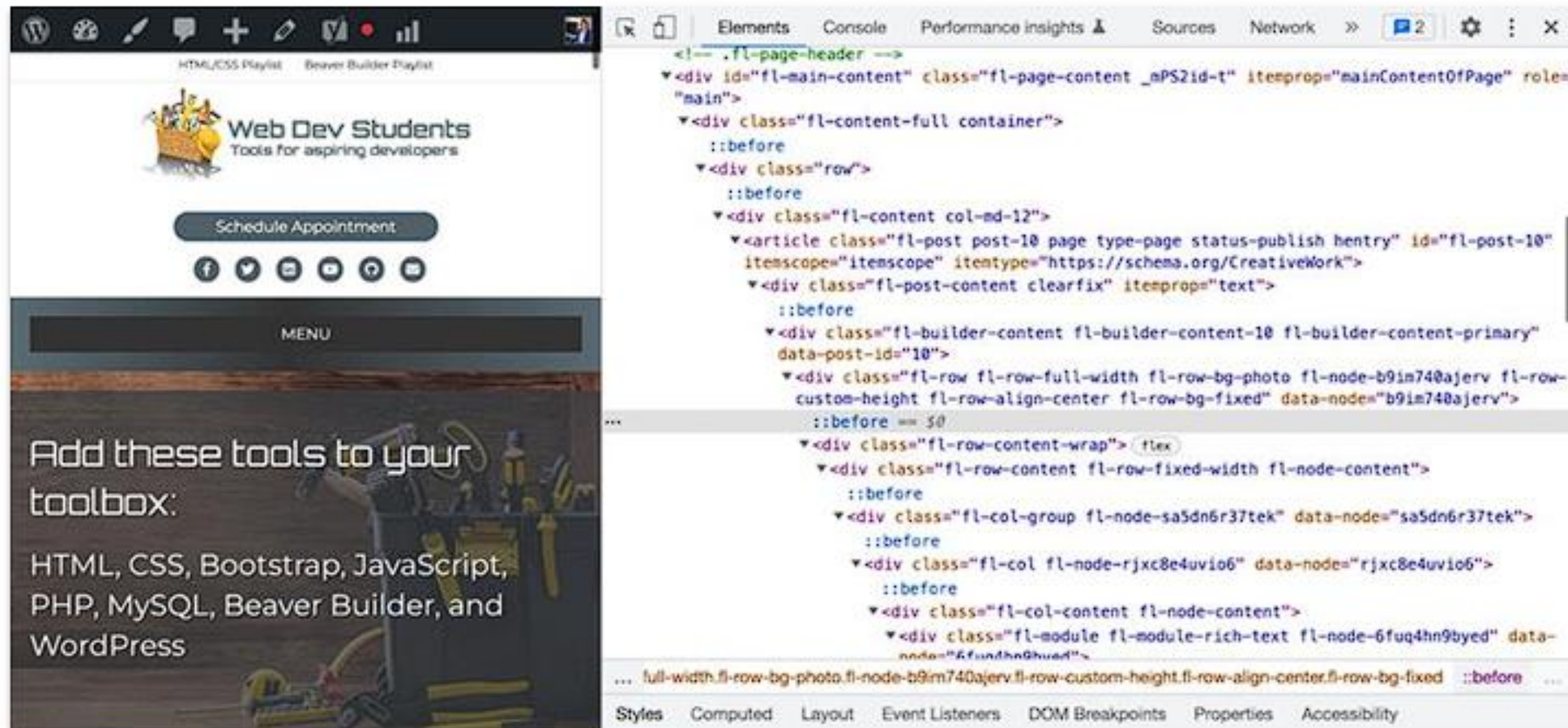
- a block element, meaning that it takes all available width, and comes with line breaks before and after.
- Used as a container.
- It's used to group content so it can be easily styled using the class or id attributes, marking a section of a document as being written in a different language (using the lang attribute), and so on.

```
index.html > html
1  <!DOCTYPE html>
2  <html>
3
4  <head>
5      <title>My First Web Page</title>
6  </head>
7
8  <body>
9      <h1>Welcome <div>To My</div> Website</h1>
10     <p>This is a paragraph</p>
11     <a href="https://www.w3schools.com/html/default.asp" target="_top">Go To This Page</a>
12 </body>
13
14 </html>
```



The <div> Element is Common

Most websites use a LOT of <div> elements. You can see them in the Chrome Inspector. The website below has more than some websites. It has more because it was created with a Content Management System.



NON-SEMANTIC ELEMENTS

`<span></span>`

- an inline container used to mark up a part of a text, or a part of a document.

NON-SEMANTIC ELEMENTS

`<span></span>`

```
index.html > html > body
1  <!DOCTYPE html>
2  <html>
3
4  <head>
5    <title>My First Web Page</title>
6  </head>
7
8  <body>
9    <h1>Welcome <span>To My</span> Website</h1>
10   <p>This is a paragraph</p>
11   <a href="https://www.w3schools.com/html/default.asp" target="_top">Go To This Page</a>
12 </body>
13
14 </html>
```



HTML TABLES

HTML tables allow web developers to arrange data into rows and columns.

- `<table></table>` - defines the table
- `<tr></tr>` - defines the table row
- `<td></td>` - defines the table cell
- `<th></th>` - defines the table header

HTML FORMS

An HTML form is used to collect user input. The user input is most often sent to a server for processing.

- `<form></form>` - used to create an HTML form for user input.
- `<input></input>` - can be displayed in many ways, depending on the type attribute.
- `<label></label>` - defines a label for many form elements.

HTML FORMS `<input>` ELEMENT

<code><input type="text"></code>	Displays a single-line text input field
<code><input type="radio"></code>	Displays a radio button (for selecting one of many choices)
<code><input type="checkbox"></code>	Displays a checkbox (for selecting zero or more of many choices)
<code><input type="submit"></code>	Displays a submit button (for submitting the form)
<code><input type="button"></code>	Displays a clickable button

THE END

THANK YOU FOR
LISTENING